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Translation of the original instructions



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Notes

Translation of the original instructions



1 Error messages

This section informs you about the error messages displayed by the TP1200.

Each fault is described with its type of error, its alarm text, its possible cause and the relevant measures for rectification.

If there is any fault, error or problem that cannot be rectified using the tables below, the authorised personnel is requested to notify the appropriate service personnel or the customer service of the manufacturer.



Operating steps and sequences on the TP1200 and further information on the visualisation on the TP1200 can be found described in the section Description – Annex Visualisation in this technical documentation.



1.1 Fault types and numbers

Types of faults

Fault with EMERGENCY STOP

A *fault with EMERGENCY STOP (MNA)* stops all active filling processes immediately (*EMERGENCY STOP* and *START OFF*).

The active filling processes are not ended properly and the *START OFF* process error is generated.

A *fault with EMERGENCY STOP (MNA)* endangers the safety of the authorised personnel and the test station.

Fault with process stop

A *fault with process stop (MPS)* stops the active filling process immediately (*START OFF*).

The active filling process is not ended properly and the *START OFF* process error is generated.

A *fault with process stop (MPS)* endangers the safety of the authorised personnel and the test station.

Fault without process stop

A *Fault without process stop (OPS)* does not stop the active filling process. The active filling process is completed. Another filling process cannot be started.

Warning

A *warning (WAR)* has no influence on the active filling process.

A warning indicates possible difficulties/irregularities in the station to the authorised personnel.

NOTE

Faults and warnings can cause process errors, including during filling and test processes.

A process error terminates the process.

A process fault forces the filling result "NOK".



Fault message numbers

The fault message number in the PLC encodes the fault for each medium.

First position

This position encodes the source of the fault.

- 0 = *CONTROL CABINET / BASE UNIT A / PUMP STATION A*
- 1 = *CONSOLE A*
- 2 = *BASE UNIT B / PUMP STATION B*
- 3 = *CONSOLE B*
- 4 = *BASE UNIT C / PUMP STATION C*
- 5 = *CONSOLE C*
- 6 = *BASE UNIT D / PUMP STATION D*
- 7 = *CONSOLE D*
- 8 = *RESERVE*
- 9 = *RESERVE*

Second position

This position encodes the filling medium.

- 0 = General fault message
- 1 = *BRAKE / CLUTCH*
- 2 = *RADIATOR (glycol)*
- 3 = *PAS*
- 4 = *UREA / AIR / PAG OIL*
- 5 = *WINDSCREEN WASHER (isopropanol)*
- 6 = *R1234yf / OIL*
- 7 = *FUEL / R744 / WATER*
- 8 = *R134a*
- 9 = ...



Third and fourth position

These two sequentially numbered positions encode the fault. Functionally identical fault messages have the same number sequence regardless of the medium.

Example

Fault message number 0120

- First position: 0 = *BASE UNIT <...>*
- Second position: 1 = *BRAKE*
- Third/fourth position 20 = *TANK OVERFILLED*

0120 = *FAULT BASE UNIT <...> BRAKE – TANK OVERFILLED*

Variables

Variables are used in the fault message text. The value of the variables is a combination of the fault message number plus the corresponding filling medium.

- *<o>* = Location of the fault message
 - *BASE UNIT <...>*
 - *CONSOLE <...>*
 - *PUMP STATION <...>*



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- *<m>* = Medium of the fault message
 - *BRAKE / CLUTCH*
 - *RADIATOR*
 - *PAS*
 - *UREA / AIR / PAG OIL*
 - *WINDSCREEN WASHER*
 - *R1234yf / OIL*
 - *FUEL / R744 / WATER*
 - *R134a*
- *<n>* = Sequential number
- *<s>* = Filling medium
 - *ALCOHOL*
 - *PETROL*
 - *BRAKE FLUID*
 - *DIESEL*
 - *ETHANOL*
 - *MIXTURE* (for radiator / windscreen washer)
 - *GLYCOL*
 - *UREA*
 - *ISOPROPANOL*
 - *AIR*
 - *OIL*
 - *PAG OIL*
 - *R1234yf*
 - *R134a*
 - *R744*
 - *PAS OIL*
 - *WATER*



Example 1

1x57 - WARNING PUMP STATION <o> <m> – BARREL 1 EMPTY - <s>

- *1257 - WARNING PUMP STATION A RADIATOR – BARREL 1 EMPTY - MIXTURE*
- *1557 - WARNING PUMP STATION A WINDSCREEN WASHER – BARREL 1 EMPTY - MIXTURE*

Example 2

1x02 - WARNING BASE UNIT <o> <m> – MOTOR PROTECTION SWITCH BARREL PUMP <s>

- *1102 - WARNING BASE UNIT A BRAKE – MOTOR PROTECTION SWITCH BARREL PUMP BRAKE FLUID*
- *1202 - WARNING BASE UNIT A RADIATOR – MOTOR PROTECTION SWITCH BARREL PUMP MIXTURE*
- *1302 - WARNING BASE UNIT A PAS – MOTOR PROTECTION SWITCH BARREL PUMP PAS OIL*
- *1502 - WARNING BASE UNIT A WINDSCREEN WASHER – MOTOR PROTECTION SWITCH BARREL PUMP MIXTURE*



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1.2 General

This section informs you about the error messages valid for the entire station, regardless of the medium.

No.	Type	Fault message	Possible cause	Remedy
Control cabinet				
0001	MNA	<i>FAULT CONTROL CABINET – EMERGENCY STOP DEVICE</i>	The emergency stop device has tripped without feedback from an emergency stop button (voltage interruption in the emergency stop circuit e.g. broken wire)	- Rectify cause for the voltage interruption (e.g. wire break) - Acknowledge the emergency stop - Switch on the line
0002	MNA	<i>FAULT CONTROL CABINET – EMERGENCY STOP BUTTON</i>	EMERGENCY STOP button at the control cabinet has been pressed	- Rectify the cause of the emergency stop - Unlock the EMERGENCY STOP button - Acknowledge the emergency stop - Switch on the line
0009	MPS	<i>FAULT CONTROL CABINET – FUSES 24VDC</i>	- Short circuit - Current consumption too high	- Check devices for overcurrent / function - Reset the fuse



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No.	Type	Fault message	Possible cause	Remedy
0017	WAR	<i>WARNING CONTROL CABINET - MOTOR PROTECTION COOLING DEVICE</i>	• Short circuit • Current consumption too high • The motor protection switch is set incorrectly	• Check devices for overcurrent / function • Set motor protection switch correctly • Connect the motor circuit breakers
0018	WAR	<i>WARNING CONTROL CABINET – ERROR COOLING DEVICE</i>	• Defective device • Defective wire • Cable not plugged in	• Check the device's function and replace if necessary • Check the connection of the unit • Check the cable and replace if necessary
0019	WAR	<i>WARNING CONTROL CABINET – KEYSWITCH OVERRIDE LINE STOP</i>	• Bridged safety signals for the conveying equipment – active / switched on key switch <i>OVERRIDE LINE STOP</i>	• <i>OVERRIDE LINE STOP</i> deactivate / switch off • Switch on the line
0020	WAR	<i>WARNING CONTROL CABINET – SOFTWARE LINESTOP</i>	Pending signal <i>SOFTWARE LINE STOP</i> at the conveying equipment • At least one console has reached the <i>END OF TRACK</i> sensor with at least one adapter that is not deposited • Defective sensor <i>END OF TRACK</i>	• Wait for process end of all active processes • Return the adapter to its holster • Move console into home position by hand if necessary • Check the sensor's function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
0021	WAR	<i>WARNING CONTROL CABINET - ANALOG INPUT LINE SPEED</i>	• Broken wire • Defective sensor • Defective analogue input module	• Check the sensor line • Check the sensor's function and replace if necessary • Check the analogue input module's function and replace if necessary
0022	WAR	<i>WARNING CONTROL CABINET - SURGE ARRESTER</i>	The overvoltage protection has triggered. • Overvoltage in the network • Defective overvoltage protection device	• Check function of overvoltage protection device and replace if necessary
0026	WAR	<i>WARNING CONTROL CABINET – SWITCH</i>	• Continuously faulty switch • Faulty self test • Missing power supply • Faulty link status of at least one port	• Check the switch's function and replace if necessary • Perform a self test • Check power supply • Check link statuses
0029	MPS	<i>FAULT CONTROL CABINET - BUS FAULT DISTRIBUTED I/O</i>	• Interrupted bus connection to the bus module • Incorrect addressing/parameterisation of the bus module • Defective bus module	• Check the bus connection • Check addressing / parameterisation and adapt if necessary • Check the bus module's function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
0031	WAR	<i>WARNING CONTROL CABINET – BUS FAULT SWITCH</i>	- Interrupted bus connection to the switch - Incorrect addressing / parameterisation of the switch - Defective switch	- Check the bus connection - Check addressing / parameterisation and adapt if necessary - Check the switch's function and replace if necessary
Console A				
1001	MNA	<i>FAULT CONSOLE <0> – EMERGENCY STOP BUTTON</i>	EMERGENCY STOP button pressed on the console <0>	- Rectify the cause of the emergency stop - Unlock the EMERGENCY STOP button - Acknowledge the emergency stop - Switch on the line
1002	WAR	<i>WARNING CONSOLE <0> – FUSE PRINTER</i>	- Short circuit - Current consumption too high	- Check devices for overcurrent / function - Reset the fuse
1003	OPS	<i>FAULT CONSOLE <0> – MOTOR PROTECTION DRIVE</i>	- Short circuit - Current consumption too high - The motor protection switch is set incorrectly	- Check devices for overcurrent / function - Set motor protection switch correctly - Connect the motor circuit breakers



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No.	Type	Fault message	Possible cause	Remedy
1004	OPS	<i>FAULT CONSOLE <0> – ADAPTER BREAK PNOZ</i>	With triggered PNOZ without triggered direction sensor <i>ADAPTER BREAK OFF FRONT</i> or <i>ADAPTER BREAK OFF REAR</i> - Defective PNOZ - Defective cable direction sensor - Defective direction sensor	- Check PNOZ function and replace if necessary - Check the cable and replace if necessary - Check the sensor's function and replace if necessary
1005	OPS	<i>FAULT CONSOLE <0> – ADAPTER BREAK IN FRONT</i>	- The adapter hose is tensioned too much - Defective wire - Defective sensor	- Loosen the adapter hose - Check the cable and replace if necessary - Check the sensor's function and replace if necessary
1006	OPS	<i>FAULT CONSOLE <0> – ADAPTER BREAK BACK</i>	- The adapter hose is tensioned too much - Defective wire - Defective sensor	- Loosen the adapter hose - Check the cable and replace if necessary - Check the sensor's function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
1007	OPS	<i>FAULT CONSOLE <0> PLAUSIBILITY ADAPTER BREAK (ACTIVATE BOTH FOR RESET)</i>	For safety reasons, the adapter break-away has a two-channel mechanism. It is monitored that both channels always switch at the same time. - Both channels did not switch at the same time. - Defective cable - Defective switch	To be able to acknowledge the fault, both switches must have switched at the same time at least once. - Check switching behaviour of the channels and restore if necessary - Check the cable and replace if necessary - Check functionality of the switch and replace if necessary
1009	OPS	<i>FAULT CONSOLE <0> – SAFETY BAR</i>	- Safety bar for bumper has tripped - Defective wire - Defective safety bar	- Remove the triggering object - Check the cable and replace if necessary - Check function of the safety bar and replace if necessary - Acknowledge safety bar
1010	OPS	<i>FAULT CONSOLE <0> – OVERRUN HOMEPOSITION</i>	- The console had overridden the sensor <i>OVERRUN HOME POSITION</i> - Defective sensor <i>OVERRUN HOME POSITION</i> - Defective wire - Defective limit switch	- Move console free from the limit switch - Check the cable and replace if necessary - Check the sensor's function and replace if necessary - Check limit switch function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
1011	OPS	<i>FAULT CONSOLE <o> – OVERRUN ENDPOSITION</i>	- The console had overridden the sensor <i>OVERRUN END POSITION</i> - Defective sensor <i>OVERRUN END POSITION</i> - Defective wire - Defective limit switch	- Move console free from the limit switch - Check the cable and replace if necessary - Check the sensor's function and replace if necessary - Check limit switch function and replace if necessary
1012	OPS	<i>FAULT CONSOLE <o> – CONVERTER DRIVE</i>	- Faulty converter - Overload (sluggish drive) - Defective converter - Defective external fan (motor)	- Read detailed fault number on converter display and look up its meaning in the technical documentation of the converter - Check ease of movement of the drive. - Check converter function, replace if necessary - Check the function of the external fan (motor) and replace if necessary
1014	OPS	<i>FAULT CONSOLE <o> – PLAUSIBILITY SYNCHRONISATION</i>	Undefined signal status at the synchronisation sensors - Defective wire - Defective sensor	- Check the cable and replace if necessary - Check the sensor's function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
1017	MPS	<i>FAULT CONSOLE <0> – BUS FAULT CONVERTER DRIVE</i>	- Interrupted bus connection to the bus module - Incorrect addressing/parameterisation of the bus module - Defective bus module	- Check the bus connection - Check addressing / parameterisation and adapt if necessary - Check the bus module's function and replace if necessary
1018	WAR	<i>WARNING CONSOLE <0> – BUS FAULT PANEL</i>	- Interrupted bus connection to the control panel - Incorrect addressing/parameterisation of the bus module - Defective control panel	- Check the bus connection - Check addressing / parameterisation and adapt if necessary - Check the control panel's function and replace if necessary
1019	MPS	<i>FAULT CONSOLE <0> – BUS FAULT DISTRIBUTED I/O</i>	- Interrupted bus connection to the bus module - Incorrect addressing/parameterisation of the bus module - Defective bus module	- Check the bus connection - Check addressing / parameterisation and adapt if necessary - Check the bus module's function and replace if necessary
1021	WAR	<i>WARNING CONSOLE <0> – BUS FAULT PANEL</i>	- Interrupted bus connection to the bus module - Incorrect addressing / parameterisation of the bus module - Defective bus module	- Check the bus connection - Check addressing / parameterisation and adapt if necessary - Check the bus module's function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
1025	OPS	<i>CONSOLE FAULT <0> – ANALOG SENSOR CONSOLE POSITION</i>	• Broken wire • Defective sensor • Defective analog input module	• Check the sensor line • Check the function of the sensor and replace if necessary • Check the analog input module's function and replace if necessary
1032	WAR	<i>WARNING CONSOLE <0> - SWITCH</i>	• Continuously faulty switch • Faulty self-test • Missing power supply • Faulty link status of at least one port	• Check the switch's function and replace if necessary • Perform a self-test • Check power supply • Check link statuses

Tab. 1: Error messages – Station – Fault general



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1.3 Filling medium *REFRIGERANT R134a*

This section informs you about the media-specific fault messages.

No.	Type	Fault message	Possible cause	Remedy
Base unit				
0810	MPS	<i>FAULT BASE UNIT <o> <m> – BUS FAULT VALVE TERMINAL</i>	• Interrupted bus connection to the bus module • Incorrect addressing/parameterisation of the bus module • Defective bus module	• Check the bus connection • Check addressing / parameterisation and adapt if necessary • Check the bus module's function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
0819	OPS	<i>FAULT BASE UNIT <o> <m> - ACCU EMPTY</i>	The fill pressure in the bladder accumulator lies below the <i>EMPTY</i> limit value (no new filling process possible) • Empty filling medium bottle • Defective supply line for filling medium • Closed ball valve in the supply line for filling medium • Clogged filter dryer • Defective filling pressure sensor • Leak in the bladder accumulator • Wrong <i>EMPTY</i> limit value has been set	• Exchange bottle and acknowledge • Check supply line of filling medium • Open all ball valves in the supply line of the filling medium • Clean filter dryer (cartridge) or replace if necessary • Check the function of the sensor and replace if necessary • Check bladder accumulator for leaks and fix if necessary • Check the <i>EMPTY</i> limit value and adapt if necessary
0820	OPS	<i>FAULT BASE UNIT <o> <m> – ACCU OVERFILLED</i>	Fill pressure in the bladder accumulator lies above the limit value <i>OVERFILLED</i> • Defective feeding pump • Defective filling pressure sensor • Defective valve in the supply line for filling medium	• Reduce pressure in the bladder accumulator to below the limit value (fill test bottle in the <i>manual mode</i>) • Check the function of the feeding pump and replace if necessary • Check the function of the sensor and replace if necessary. • Check the function of the valve and replace if necessary.



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No.	Type	Fault message	Possible cause	Remedy
0825	WAR	<i>WARNING BASE UNIT <o> AC - TIME OUT REFILLING ACCU</i>	The runtime starts with the beginning of the refilling process. The switching point <i>MAXIMUM</i> of the fill pressure sensor is not reached in the prescribed time • Closed ball valve in the supply line for filling medium • Clogged filter dryer • Pressure in the filling medium supply line is too low • Too warm feeding pump • Too low control air pressure of the feeding pump • The set run-time is too short • Defective valve • Defective filling pressure sensor	• Open all ball valves in the supply line of the filling medium • Clean filter dryer (cartridge) or replace if necessary • Ensure input pressure of the filling medium (5 bar) • Increase the runtime • Check the function of the pump cooling and replace if necessary • Ensure the control air pressure of the feeding pump • Check the valve's function and replace if necessary • Check the sensor's function and replace if necessary
0826	WAR	<i>WARNING BASE UNIT <o> <m> – PRESSURE SUPPLY PIPE</i>	• Pressure in the filling medium supply line is too low • The pressure switch is set incorrectly • Defective pressure switch	• Increase the pressure in the filling medium supply line • Properly set the pressure switch • Check function of the pressure switch and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
0833	OPS	<i>FAULT BASE UNIT <o> AC – ANALOG SENSOR FILLING PRESSURE BLADDER ACCU</i>	• Broken wire • Defective sensor • Defective analog module • Fill pressure sensor provides an analog value less or equal to zero (no reliable switch-off is ensured during refilling).	• Check the sensor line. • Check the function of the sensor and replace if necessary. • Check the analog module's function and replace if necessary. • Check pressure in the bladder accumulator
0857	WAR	<i>WARNING PUMP STATION <o> <m> – BOTTLE 1 EMPTY – <m></i>	• Empty filling medium bottle • Closed ball valve in the supply line for filling medium • Defective pressure switch • The pressure switch is set incorrectly	• Exchange bottle and acknowledge • Open all ball valves in the supply line of the filling medium • Check the function of the pressure switch and replace if necessary • Set the pressure switch correctly
0859	WAR	<i>WARNING PUMP STATION <o> <m> – BOTTLE 1 EMPTY - NITROGEN</i>	• Empty bottle • Closed ball valve in the supply line for filling medium • Defective pressure switch • The pressure switch is set incorrectly	• Exchange bottle and acknowledge • Open all ball valves in the supply line of the filling medium • Check function of the pressure switch and replace if necessary • Properly set the pressure switch



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No.	Type	Fault message	Possible cause	Remedy
0860	WAR	<i>WARNING PUMP STATION <o> <m> – BOTTLE 2 EMPTY - NITROGEN</i>	• Empty bottle • Closed ball valve in the supply line for filling medium • Defective pressure switch • The pressure switch is set incorrectly	• Exchange bottle and acknowledge • Open all ball valves in the supply line of the filling medium • Check function of the pressure switch and replace if necessary • Properly set the pressure switch
0880	OPS	<i>FAULT BASE UNIT <o> <m> – VALVE TERMINAL - COLLECTED ERROR</i>	See dialog window <i>DIAGNOSTICS</i> on the TP1200	• Check function of the Festo terminal
Console part				
1801	MPS	<i>FAULT CONSOLE <o> <m> – COMPRESSED AIR SUPPLY</i>	• Compressed air supply at the pneumatic maintenance unit is switched off • Inadequate pressure in the compressed air supply • Defective pressure switch • The pressure switch is set incorrectly	• Switch on the compressed air supply at the pneumatic maintenance unit • Establish the compressed air supply • Check function of the pressure switch and replace if necessary • Properly set the pressure switch



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No.	Type	Fault message	Possible cause	Remedy
1802	MPS	<i>FAULT CONSOLE <o> <m> – BUS FAULT VALVE TERMINAL</i>	- Interrupted bus connection to the bus module - Incorrect addressing/parameterisation of the bus module - Defective bus module	- Check the bus connection - Check addressing / parameterisation and adapt if necessary - Check the bus module's function and replace if necessary
1803	MPS	<i>FAULT CONSOLE <o> <m> – MO- TOR PROTECTION VACUUM PUMP</i>	- Short circuit - Current consumption too high - The motor protection switch is set incorrectly	- Check devices for overcurrent / function - Set motor protection switch correctly - Connect the motor circuit breakers
1804	OPS	<i>FAULT CONSOLE <o> <m> – NO FILLING PRESSURE <m></i>	- Pressure in the bladder accumulator is too low - Closed ball valve in the filling line for filling medium - The pressure switch is set incorrectly - Defective pressure switch - Defective valve	- Refill bladder accumulator - Open all ball valves in the filling line of the filling medium - Set the pressure switch correctly - Check the function of the pressure switch and replace if necessary - Check the function of the valve and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
1805	OPS	<i>FAULT CONSOLE <o> <m> – NO PRESSURE FOR PRESSURE TEST</i>	• Inadequate pressure in the supply test medium • Closed ball valve in the pressure test line for test medium • The pressure switch is set incorrectly • Defective pressure switch • Defective valve	• Check and acknowledge test medium supply • Open all ball valves in the pressure test line of the test medium • Set the pressure switch correctly • Check the function of the pressure switch and replace if necessary • Check the function of the valve and replace if necessary.
1806	OPS	<i>FAULT CONSOLE <o> <m> – COUNTER FILLING WEIGHT <m></i>	• Broken wire • Clogged volume counter • Defective volume counter • Defective counter module	• Check the sensor line • Check the function of the volume counter and clean or replace if necessary • Check the function of the counter module and replace if necessary.
1810	OPS	<i>FAULT CONSOLE <o> <m> – ANALOG SENSOR VACUUM</i>	• Broken wire • Defective sensor • Defective analogue input module	• Check the sensor line • Check the function of the sensor and replace if necessary • Check the analogue input module's function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
1811	OPS	<i>FAULT CONSOLE <o> <m> – ANALOG SENSOR PRESSURE TEST</i>	• Broken wire • Defective sensor • Defective analog input module	• Check the sensor line. • Check the function of the sensor and replace if necessary. • Check the analog input module's function and replace if necessary.
1814	OPS	<i>FAULT CONSOLE <o> <m> – ANALOG SENSOR FILLING PRESSURE CONSOLE</i>	• Broken wire • Defective sensor • Defective analogue input module • Sensor value deviates too much from the comparison sensors.	• Check the sensor line • Check the sensor's function and replace if necessary • Check the analogue input module's function and replace if necessary • Calibrate the sensor
1817	OPS	<i>FAULT CONSOLE <o> <m> – ADAPTER CLAMPED IN HOLSTER</i>	• Adapter rest clamped in the holster (PLC step 9-0 WAITING FOR ADAPTER STOP) • Defective STOP button on the adapter • Defective sensor <i>HOLSTER</i>	• End the PLC step sequence (Press the STOP button) • Check function of the STOP button and replace if necessary • Check the sensor's function and replace if necessary



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No.	Type	Fault message	Possible cause	Remedy
1818	OPS	<i>FAULT CONSOLE <o> <m> – COUNTER STANDSTILL DURING FILLING <m></i>	A counter standstill was detected during the volume filling • Closed ball valve in the filling line for filling medium • Clogged volume counter • Defective volume counter • Defective counter module • Defective fill valve • Clogged filter	• Open all ball valves in the filling line of the filling medium. • Check the function of the volume counter and clean or replace if necessary. • Check the function of the counter module and replace if necessary. • Check the valve's function and replace if necessary. • Clean filter or replace if necessary.
1820	OPS	<i>FAULT CONSOLE <o> <m> – PROMASS NOT READY – AIR IN SYSTEM</i>	Missing input <i>PROMASS READY</i> • Broken wire • PROMASS defective • PROMASS recognises air in the system and generates internal error. If this fault occurs during filling, the process error NOK FILLING VOLUME will be generated after filling has been completed.	• Check the sensor line • Check PROMASS function and replace if necessary. • Read error number on PROMASS display and search in the PROMASS manual. • Remove air from the system by filling test bottles.



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No.	Type	Fault message	Possible cause	Remedy
1822	OPS	<i>FAULT CONSOLE <o> <m> – CLEANING HP-ADAPTER AFTER PAG OIL FILLING (BLOW OUT IN FILTER)</i>	Due to the option "ONLY FILLING PAG OIL", the HP adapter has been contaminated with PAG oil, which must under no circumstances get into the vacuum pump. Therefore the HP adapter first has to be cleaned. To this end, it is purged with refrigerant. "ONLY FILLING PAG OIL" option has been deselected	- Open the door of the hydraulic cabinet - Activate the "OVERRIDE DOOR" key switch - Place the HP adapter on the appropriate filter with nipple and press the START button to clamp it - The HP adapter cleans itself - Press the STOP button to unclamp the HP adapter and place it in the holster - Close the door of the hydraulic cabinet - Deactivate "OVERRIDE DOOR" key switch - Press the RESET button to reset the fault
1824	MPS	<i>FAULT CONSOLE <o> <m> – OVERLOAD RELAY VACUUM PUMP</i>	- Short circuit - Current consumption too high - Fault in the network symmetry	- Check vacuum pump for overload. - Check phases of the vacuum pump. - Reset overload relay.



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No.	Type	Fault message	Possible cause	Remedy
1825	OPS	<i>FAULT CONSOLE <o> <m> – ADAPTER SEALING - ADAPTER NOT CLAMPED CORRECTLY</i>	The adapter seal test was unsuccessful. The permissible vacuum limit values were exceeded. • Adapter clamped onto free space • Adapter not correctly positioned when clamped • Adapter damaged • Vehicle connections damaged • Fault in the vacuum system of the station (vacuum pump, vacuum valve, vacuum sensor) • Evacuation time, adaptation test, too low • Adaptation test leak limit value too low	• Always attach adapter onto vehicle or test bottle. • Always attach adapter perpendicular to the connections with mild pressure. • Check adapter for damage • Check vehicle connections for damage/leaks. • Check station vacuum system. • Increase evacuation time of the adaptation test. • Increase adaptation test leak limit value
1826	OPS	<i>FAULT CONSOLE <o> <m> – TEST CAR VALVES - LEAKY</i>	• At least one valve (HP/LP) on the system of the vehicle leaky	• Acknowledge fault, unclamp adapter and seal vehicle valve • If necessary, extract R1234yf from the vehicle and check valves.



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No.	Type	Fault message	Possible cause	Remedy
1827	OPS	<i>FAULT CONSOLE <o> <m> – SYSTEM VACUUM TO BAD</i>	The vacuum in the vacuum system has exceeded the permissible limit value for adapter in the holster. • Leaky vacuum system • A vacuum valve does not open. • Defective vacuum pump • Defective vacuum sensor	• Check whether the vacuum system is leak-tight • Check the valve's function and replace if necessary • Check function of vacuum pump and replace if necessary • Check the sensor's function and replace if necessary
1836	OPS	<i>FAULT CONSOLE <o> <m> – ADAPTER VALVE V8560 LEAKY</i>	• Leaky adapter valve V8560 • Defective pressure sensor • Defective analog module for pressure sensor	• Replace valve • Check the function of the sensor and replace if necessary • Check the function of the analog module and replace if necessary
1848	OPS	<i>FAULT CONSOLE <o> <m> – VALVE TERMINAL - COLLECTED ERROR</i>	See dialog window <i>DIAGNOSTICS</i> on the TP1200	• Check function of the Festo terminal

Tab. 2: Fault messages – REFRIGERANT filling medium R134a



04-01 Troubleshooting

2 Process errors

Process errors occur during the filling or test process.

Process errors are caused by the vehicle system or the station.

Process faults are generated when the step-forward conditions in a step sequence are not met.

Process errors are only formed in AUTOMATIC MODE.

Process errors are always reset in the other modes.

Process fault are generated as they occur and interrupt the filling or test process immediately.

The process fault is active that interrupts the filling or test process.

Process errors indicate at which step of the automatic step sequence the filling or test process was interrupted.



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2.1 General

This section informs you about the process error valid for the entire station, regardless of the medium.

PLC step	Error message	Description	Possible cause	Remedy
Each step	<i>NOK: ADAPTER STOP</i>	The STOP button is pressed by the operator during the filling process	• STOP button pressed by the operator • Faulty pulses from STOP button	• Release STOP button • Check the functionality of the STOP button and replace if necessary
Each step	<i>NOK: START OFF</i>	Station / process is switched off during the filling or test process	• by the operator pressing the button: <i>START OFF</i> • Station fault with <i>START OFF</i>	• Rectify all station faults • Restart the station

Tab. 3: Process errors – Process errors station



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2.2 Filling medium *REFRIGERANT R134a*

This section describes the filling process-dependent process errors.

PLC step	Error message	Description	Possible cause	Remedy
3-1	<i>NOK: PRESSURE BUILD-UP</i>	Specified nominal pressure for the pressure test build-up not reached within the nominal time	• Leaky vehicle system • Leaky vacuum system • Defective pressure test sensor • Defective valve • Defective proportional valve • Defective pressure regulator • Setpoint too high • Nominal time too short • <i>OFFSET</i> too low for the pressure build-up of the proportional valve	• Check whether the vehicle system is leak-tight and stop leak if necessary • Check whether the vacuum system is leak-tight and stop leak if necessary • Check the sensor's function and replace if necessary • Check the valve's function and replace if necessary • Check function of the pressure switch and replace if necessary • Check setpoint and adapt if necessary • Check nominal time and adapt if necessary • Check <i>OFFSET</i> and adjust if necessary



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PLC step	Error message	Description	Possible cause	Remedy
3-11	<i>NOK: PRESSURE BUILD-UP SETTLING</i>	Specified nominal pressure for the pressure test build-up not reached within the stabilisation time	• Leaky vehicle system • Leaky vacuum system • Defective pressure test sensor • Defective valve • Too short nominal time <i>PRESSURE BUILD-UP</i> • <i>OFFSET</i> too low for the pressure build-up of the proportional valve	• Check whether the vehicle system is leak-tight and stop leak if necessary • Check whether the vacuum system is leak-tight and stop leak if necessary • Check the sensor's function and replace if necessary • Check the valve's function and replace if necessary • Check nominal time and adapt if necessary • Check <i>OFFSET</i> and adjust if necessary
3-2	<i>NOK: PRESSURE LEAK TEST</i>	Specified maximum pressure leakage value exceeded within the leak test time	• Leaky vehicle system • Leaky vacuum system • Setpoint too low • Too short stabilisation time (during PLC step 3-11)	• Check whether the vehicle system is leak-tight and stop leak if necessary • Check whether the vacuum system is leak-tight and stop leak if necessary • Check setpoint and adapt if necessary • Check stabilisation time and adjust if necessary



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PLC step	Error message	Description	Possible cause	Remedy
3-3	<i>NOK: PRESSURE RELIEF</i>	Specified pressure equalization (ambient pressure with tolerances) not reached within the nominal time	- Defective valve - Setpoint too low - Nominal time too short	- Check the valve's function and replace if necessary - Check setpoint and adapt if necessary - Check nominal time and adapt if necessary
4-0	<i>NOK: COARSE VACUUM</i>	Specified setpoint for the coarse vacuum not reached within the nominal time	- Leaky vehicle system - Leaky vacuum system - Defective vacuum sensor - Defective valve - Defective vacuum pump - Setpoint too low - Nominal time too short	- Check whether the vehicle system is leak-tight and stop leak if necessary - Check whether the vacuum system is leak-tight and stop leak if necessary - Check the sensor's function and replace if necessary - Check the valve's function and replace if necessary - Check function of vacuum pump and replace if necessary - Check setpoint and adapt if necessary - Check nominal time and adapt if necessary



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PLC step	Error message	Description	Possible cause	Remedy
4-0	<i>NOK: VACUUM TOO FAST</i>	A suitable vacuum value has been reached too quickly in the coarse vacuum	• "Vacuum/Aeration" adapter valve defective • "Release vacuum sensor" valve defective • Defective vacuum sensor	• Check the function of the adapter valve and replace if necessary • Check the function of the valve and replace if necessary • Check the function of the sensor and replace if necessary



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PLC step	Error message	Description	Possible cause	Remedy
4-1	<i>NOK: FINE VACUUM</i>	Specified setpoint for the fine vacuum not reached within the set time	• Leaky vehicle system • Leaky vacuum system • Defective vacuum sensor • Defective valve • Defective vacuum pump • Setpoint too low • Nominal time too short	• Check whether the vehicle system is leak-tight and stop leak if necessary • Check whether the vacuum system is leak-tight and stop leak if necessary • Check the sensor's function and replace if necessary • Check the valve's function and replace if necessary • Check function of vacuum pump and replace if necessary • Check setpoint and adapt if necessary • Check nominal time and adapt if necessary



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PLC step	Error message	Description	Possible cause	Remedy
4-2	<i>NOK: VACUUM LEAK TEST</i>	Specified maximum vacuum leakage value exceeded within the leak test time	- Leaky vehicle system - Leaky vacuum system - Setpoint too low - Evacuation time too short (during PLC step 4-0 and 4-1)	- Check whether the vehicle system is leak-tight and stop leak if necessary - Check whether the vacuum system is leak-tight and stop leak if necessary - Check setpoint and adapt if necessary - Check evacuation time and adjust if necessary
4-3	<i>NOK: POST VACUUM</i>	Specified setpoint for the post-vacuum not reached within the set time	- Setpoint too low - Nominal time too short	- Check setpoint and adapt if necessary - Check nominal time and adapt if necessary
5-0	<i>NOK: ZEROISE COUNTER</i>	Actual value of the counter at the end of the "Reset Counter" step (running time=2s) is more than 10 ml	- Leaky filling line for filling medium - Defective volume counter - Defective counter module	- Check whether the filling line of the filling medium is leak-tight and stop leak if necessary - Check function of volume counter and replace if necessary - Check counter module's function and replace if necessary



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PLC step	Error message	Description	Possible cause	Remedy
5-1 to 5-6	<i>NOK: COUNTER STAND- STILL</i>	Counter standstill detected during the PLC step <i>DOSING / FILLING</i> .	• Closed ball valve in the filling line for filling medium • Defective filling pump • Defective volume counter • Defective counter module • Defective valve • Clogged filter	• Open all ball valves in the filling line of the filling medium. • Check function of filling pump and replace if necessary. • Check the function of the volume counter and replace if necessary. • Check the function of the counter module and replace if necessary. • Check the valve's function and replace if necessary. • Clean filter or replace if necessary.



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PLC step	Error message	Description	Possible cause	Remedy
5-3	NOK: FILLING TIME	Specified nominal volume not reached within the permitted time (only during volume filling)	• Closed ball valve in the filling line for filling medium • Defective volume counter • Defective counter module • Defective valve • Clogged filter • Setpoint too high • Nominal time too short	• Open all ball valves in the filling line of the filling medium • Check function of volume counter and replace if necessary • Check counter module's function and replace if necessary • Check the valve's function and replace if necessary • Clean filter or replace if necessary • Check setpoint and adapt if necessary • Check nominal time and change if necessary
5-3	NOK: ADAPTER VALVE	Registered pressure increase on the vacuum side at the pressure test sensor during the AC filling process	• Leaky adapter valve V8560 • Defective pressure test sensor • Wrong limit value (typically: 400 mbar relative)	• Check the valve's function and replace if necessary • Check the sensor's function and replace if necessary • Check limit values and adapt if necessary



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PLC step	Error message	Description	Possible cause	Remedy
6-0 / 9-0	NOK: FILLING VOLUME	Filling volume at end of filling is outside tolerance range of the nominal volume	• Already filled system filling medium • Defective volume counter • Defective counter module • Defective valve • Wrong setpoint • Setpoint too low for tolerance	• Check fill level at the system filling medium. • Check the function of the volume counter and replace if necessary. • Check the function of the counter module and replace if necessary. • Check the function of the valve and replace if necessary. • Check setpoint and adjust if necessary.

Tab. 4: Process error – Filling process AC

